

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech. in COMPUTER SCIENCE AND ENGINEERING
COURSE STRUCTURE, I & II YEAR SYLLABUS (R22 Regulations)

Applicable from AY 2023-24 Batch

I Year I Semester

S. No.	Course Code	Course	L	T	P	Credits
1.	CS1101	Matrices and Calculus	3	1	0	4
2.	CS1102	Applied Physics	3	1	0	4
3.	CS1103	Programming for Problem Solving	3	0	0	3
4.	CS1104	English for Skill Enhancement	2	0	0	2
5.	CS1105	Computer Aided Engineering Graphics	1	0	4	3
6.	CS1106	Elements of Computer Science & Engineering	0	0	2	1
7.	CS1107	Applied Physics Laboratory	0	0	3	1.5
8.	CS1108	Programming for Problem Solving Laboratory	0	0	2	1
9.	CS1109	English Language and Communication Skills Laboratory	0	0	2	1
Total			12	2	12	20.5

I Year II Semester

S. No.	Course Code	Course	L	T	P	Credits
1.	CS1201	Ordinary Differential Equations and Vector Calculus	3	1	0	4
2.	CS1202	Engineering Chemistry	3	1	0	4
3.	CS1203	Engineering Workshop	0	1	3	2.5
4.	CS1204	Basic Electrical Engineering	2	0	0	2
5.	CS1205	Electronic Devices and Circuits	2	0	0	2
6.	CS1206	Data Analysis using Excel	0	0	3	1
7.	CS1207	Python Programming Laboratory	0	1	2	2
8.	CS1208	Basic Electrical Engineering Laboratory	0	0	2	1
9.	CS1209	IT Workshop	0	0	2	1
Total			10	4	12	19.5

II YEAR I SEMESTER

S. No.	Course Code	Course Title	L	T	P	Credits
1	CS2101	Digital Electronics	3	0	0	3
2	CS2102	Data Structures	3	0	0	3
3	CS2103	Computer Oriented Statistical Methods	3	1	0	4
4	CS2104	Computer Organization and Architecture	3	0	0	3
5	CS2105	Object Oriented Programming through Java	3	0	0	3
6	CS2106	Data Structures Lab	0	0	3	1.5
7	CS2107	Object Oriented Programming through Java Lab	0	0	3	1.5
8	CS2108	Gender Sensitization Lab	0	0	2	0
9	CS2109	Skill Development Course (Data visualization- R Programming/ Power BI)	0	0	2	1
Total			15	1	10	20

II YEAR II SEMESTER

S. No.	Course Code	Course Title	L	T	P	Credits
1	CS2201	Discrete Mathematics	3	0	0	3
2	CS2202	Business Economics & Financial Analysis	3	0	0	3
3	CS2203	Operating Systems	3	0	0	3
4	CS2204	Database Management Systems	3	0	0	3
5	CS2205	Software Engineering	3	0	0	3
6	CS2206	Operating Systems Lab	0	0	2	1
7	CS2207	Database Management Systems Lab	0	0	2	1
8	CS2208	Real-time Research Project/ Societal Related Project	0	0	4	2
9	CS2209	Constitution of India	3	0	0	0
10	CS2210	Skill Development Course (Node JS/ React JS/ Django)	0	0	2	1
Total			18	0	10	20

III YEAR I SEMESTER

S. No.	Course Code	Course Title	L	T	P	Credits
1		Design and Analysis of Algorithms	3	1	0	4
2		Computer Networks	3	0	0	3
3		DevOps	3	0	0	3
4		Professional Elective-I	3	0	0	3
5		Professional Elective –II	3	0	0	3
6		Computer Networks Lab	0	0	2	1
7		DevOps Lab	0	0	2	1
8		Advanced Communication Skills Lab	0	0	2	1
9		Intellectual Property Rights	3	0	0	0
10		Skill Development Course (UI design- Flutter)	0	0	2	1
Total			18	1	8	20

III YEAR II SEMESTER

S. No.	Course Code	Course Title	L	T	P	Credits
1		Machine Learning	3	0	0	3
2		Formal Languages and Automata Theory	3	0	0	3
3		Artificial Intelligence	3	0	0	3
4		Professional Elective – III	3	0	0	3
5		Open Elective-I	3	0	0	3
6		Machine Learning Lab	0	0	2	1
7		Professional Elective-III Lab	0	0	2	1
8		Industrial Oriented Mini Project/ Internship/ Skill Development Course (Big data-Spark)	0	0	4	2
9		Environmental Science	3	0	0	0
Total			18	0	8	20

Environmental Science in III Yr II Sem Should be Registered by Lateral Entry Students Only.

IV YEAR I SEMESTER

S. No.	Course Code	Course Title	L	T	P	Credits
1		Cryptography and Network Security	3	0	0	3
2		Compiler Design	3	0	0	3
3		Professional Elective –IV	3	0	0	3
4		Professional Elective –V	3	0	0	3
5		Open Elective – II	3	0	0	3
6		Cryptography and Network Security Lab	0	0	2	1
7		Compiler Design Lab	0	0	2	1
8		Project Stage – I	0	0	6	3
		Total Credits	15	0	10	20

IV YEAR II SEMESTER

S. No.	Course Code	Course Title	L	T	P	Credits
1		Organizational Behavior	3	0	0	3
2		Professional Elective – VI	3	0	0	3
3		Open Elective – III	3	0	0	3
4		Project Stage – II including Seminar	0	0	22	9+2
		Total Credits	9	0	22	20

MC – Satisfactory/Unsatisfactory*#Skill Course - 1 credit with 2 Practical Hours****Professional Elective - I**

	Quantum Computing
	Advanced Computer Architecture
	Data Analytics
	Image Processing
	Principles of Programming Languages

Professional Elective - II

	Computer Graphics
	Embedded Systems
	Information Retrieval Systems
	Distributed Databases
	Natural Language Processing

Professional Elective - III

	Full Stack Development
	Internet of Things
	Scripting Languages
	Mobile Application Development
	Software Testing Methodologies

Courses in PE - III and PE - III Lab must be in 1-1 correspondence.**Professional Elective -IV**

	Graph Theory
	Advanced Operating Systems
	Soft Computing
	Cloud Computing
	Ad hoc & Sensor Networks

Professional Elective -V

	Advanced Algorithms
	Agile Methodology
	Robotic Process Automation
	Blockchain Technology
	Software Process & Project Management

Professional Elective – VI

	Computational Complexity
	Distributed Systems
	Deep Learning
	Human Computer Interaction
	Cyber Forensics